

Report on Participation in the “Computer Vision in Archaeology” Training School, Brno, Czech Republic (14–18 September 2026)

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Figure 1 Group photo of the participants (Photo: [ARUP-CAS](#))

This intensive course offered a brief introduction on the topic of computer vision in archaeology and proceeded with a range of applications for the detection, classification and analysis of the archaeological material. Different datasets related to various artifacts, microscopy, remote sensing, rock art and numismatic studies were provided by experienced tutors as examples for the method demonstration and the practical training of the participants.

Day 1: Intro to Computer Vision, coding with agents & Python - 14/9/2026

On the first day, the school organizers and participants were introduced. An introduction session on the fundamental computer vision concepts and applications in archaeology, was given by Petr Pajdla, highlighting the workflow, possibilities and limitations accompanied by case studies. During the session, discussions were made which aimed to build interactions among the different groups of trainees and to develop active participatory learning. On the practical part, a computer vision-language model in Google Colab - CLIP, was used for zero-shot classification. Agentic programming was introduced by Ronald Harasim to help the participants understand how to code and what to avoid when working with Agents, while Filip Hájek presented Python basics. The day ended with an ice-breaker event that was adequate for creating a more comfortable and collaborative atmosphere for the following sessions among the participants.

Day 2: Artefact Datasets & Projects' presentations - 15/9/2026

On the second day, the topics covered were artefact image datasets and image annotation. Dataset preparation and the concept of annotation, including object identification and labelling, were explained and practised using CVAT, an open-source, web-based platform for image annotation. Trying ArchaeoTag, a game for annotating archaeological artifacts was an entertaining addition as well. During the afternoon, a number of participants presented their projects and the day ended with a city tour of Brno's historic centre.

Day 3: Microscopic images dataset - 16/9/2026

On the third day, microscopic image datasets with image analysis, dataset preparation, standardization and automated segmentation were studied. Anastasia Eleftheriadou, the presenting lecturer, used datasets from stone tools use-wear analysis and sediment thin sections (micromorphology) as case studies. The practical session included provided codes to run on Google Colab for further understanding of the relevant processes. Additionally, a tour visit given by the Institute's laboratories hosts offered opportunities to see the updated equipment of the facility. During the evening, a social event encouraged further discussions among the participants.

Day 4: Satellite imagery & Rock art datasets - 17/9/2026

On the fourth day, Filip Hájek lectured in the morning on the use of computer vision on satellite imagery. The practical part covered the application of automated techniques for the detection of archaeological features on terrestrial surfaces. The afternoon session started with a presentation on a rock art case study given by Ashely Green. The participants used Label studio to perform annotations and a pre-trained YOLO segmentation model on Google Colab was used for detecting cupmarks in archaeological images. Additionally, the day also included an interesting visit to the library of the institute.

Day 5: Coins dataset & closing of the training school - 18/9/2026

On the last day of the summer school, Ronald Harasim showed the trainees how to apply computer vision workflows on coin datasets. Large language models (LLMs) and vision-language models were presented. The summer school ended with a closing session and discussion on the week's topics and future research applications.

Conclusions & impact on my doctoral research

This training school was a valuable experience that provided a solid brief introduction and recapping on the basics of computer vision in archaeology - showing several potential uses, highlighting possibilities and limitations. I had the opportunity to present my ongoing doctoral project and receive useful feedback. It gave me further motivation to continue my thesis research on rock art which incorporates these technological aspects for morphological and stylistic classification. In addition, it was a unique opportunity for networking, potential future collaborations and discussing other perspectives on machine learning in archaeology. Henceforth, I would like to express my deep gratitude to the organisers for giving me the opportunity to attend with a grant and moreover for their hospitality throughout the program!